Edge-Enhanced Cascaded MRF for SAR Image Segmentation

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***Abstract*— Markov random fields (MRFs) effectively capture local contextual information by modeling the spatial dependencies between pixels, which helps highlight details and enhances segmentation smoothness. To fully exploit MRF for synthetic aperture radar (SAR) image segmentation, we propose a novel edge-enhanced cascaded MRF (ECMRF) approach. Specifically, we introduce multiple edge-constrained filters to emphasize SAR image boundaries and provide relatively clean features. Building on this, we present a cascaded MRF framework that sequentially integrates region-level and pixel-level segmentations with feature perturbation and fusion to generate the final segmentation output. The framework comprises four key components: 1) a region-level MRF (RMRF), regulated by edge features, to achieve precise region segmentation; 2) a pixel-level MRF (PMRF) with selective label smoothing to refine edges and reduce noise clusters; 3) equal-channel feature perturbation (EC-FP) to increase feature diversity; and 4) a random probability-based feature fusion (RP-FF) scheme to merge the input features. The experimental results demonstrate that our ECMRF outperforms six state-of-the-art comparable methods, underscoring its competitive performance.**

***Index Terms*— Cascaded Markov random field (MRF), edge-constrained filters, feature perturbation, synthetic aperture radar (SAR) image.**

I. Introduction

SYNTHETIC aperture radar (SAR) imagery, a key form of remote sensing data, differs from optical images due to its distinct imaging mechanism [1], [2]. SAR systems provide high-precision observations regardless of weather or lighting conditions [3], making them crucial for both civilian [4] and military [5] applications. SAR images are widely used in target recognition [6], change detection [7], land cover classification [8], and other high-level applications [9]. However, they are highly susceptible to speckle noise, which blurs boundaries, reduces detail, and leads to over-segmentation. As a result, achieving accurate SAR image segmentation [10] remains a significant challenge.

The Markov random field (MRF) is a probabilistic graphical model rooted in Markov chains and Bayesian theory [11]. By modeling spatial dependencies between pixels, MRF enhances edge and corner features while ensuring label consistency [12], thereby improving the accuracy of image classification and segmentation [13]. In image segmentation tasks, pixel-level MRF (PMRF) constructs an objective function that integrates spatial contextual information and visual features [14]. Optimizing this function assigns optimal labels to each pixel, enabling precise segmentation. However, the inherent speckle noise in SAR images disrupts the spatial dependence modeling in MRF, causing inconsistent labels in targets, which ultimately degrade the accuracy of the segmentation [15]. To address this problem, CWNN-MRF [16] replaces traditional pooling with wavelet-constrained pooling to predict SAR image labels, achieving local smoothing through superpixel labeling while utilizing PMRF for edge enhancement. An edge description map (sketch map) is employed to replace edge regions in the superpixel map with corresponding PMRF outputs, enhancing segmentation detail while maintaining internal consistency within the region. Nevertheless, pixel-level operations incur high computational costs. To improve computational efficiency, the region-level MRF (RMRF) [17] utilizes initial segmentation blocks derived from watershed transformation as processing units. Similar to PMRF, it models spatial contextual relationships and feature information of these regions. RCC-MRF [18] constructs an RMRF objective function incorporating spatial context, deep features, and region category confidence to achieve SAR image segmentation, enhancing local smoothing capability. However, improper region partitioning can result in coarse object boundaries, potentially reducing segmentation accuracy (SA) [19]. Thus, neither PMRF nor RMRF can simultaneously achieve segmentation performance and computational efficiency.

To achieve fast and accurate segmentation, FWMRF [20] employs a fast and robust fuzzy C-means (FCM) algorithm to divide SAR images into homogeneous and heterogeneous

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areas. It then applies RMRF and PMRF to these areas separately, leveraging their strengths in region labeling and edge segmentation. However, this method relies on simple L_0 smoothing, which struggles to effectively suppress noise, and low-quality prior labels constrain its segmentation performance. While these methods have enhanced SAR image segmentation by integrating MRF, they still face challenges in handling complex boundary targets and are prone to falling into local optima. To combat this, Shang et al. [21] proposed a class-label correction method based on pixel grouping and region growing after using MRF to refine corner pixels. This approach strengthens global consistency through hierarchical image information modeling while more accurately capturing local features. Shang et al. [22] adopted a sequential strategy by introducing label-assisted region fusion after MRF processing and then reapplying MRF to refine edge region labels. Experiments have demonstrated that applying MRF at multiple stages significantly improves segmentation performance. These studies confirm that a well-designed cascaded MRF framework can effectively enhance the accuracy of SAR image segmentation.

To enhance noise suppression and segmentation robustness, it is crucial to investigate both filtering techniques and K-means clustering concurrently. Filtering methods mitigate speckle noise while preserving critical structural details, whereas K-means offers an efficient unsupervised framework [23] for generating prior labels. Shang et al. [21] applied an edge-constrained Gaussian filtering (ECGauF) to SAR images and treated the filtered results as input for K-means clustering, thereby improving the preservation of boundary details. However, relying on a single filtering approach may result in the loss of details, incomplete boundary information, and insufficient noise suppression. Inspired by genetic algorithm (GA) [24], increasing population diversity can help avoid local optima [25] and facilitate the discovery of global optimal solutions. Shang et al. [26] integrated three distinct detection operators—mean ratio, gradient, and cross correlation—for SAR edge detection, thereby enhancing the algorithm's robustness and improving edge detection accuracy. Although these methods have made notable contributions to SAR image filtering, none can simultaneously achieve detailed preservation and provide effective segmentation dynamics.

In response to these challenges and insights, we propose a novel edge-enhanced cascaded MRF (ECMRF) approach for SAR image segmentation. We first introduce multiple edge-constrained filters to extract diverse edge features and mitigate noise interference. Subsequently, we present a cascaded MRF framework that sequentially integrates region-level and pixel-level segmentations with feature perturbation and fusion to generate the final segmentation output. The framework comprises four key components.

1) Edge-regulated rapid RMRF (ERR-RMRF) utilizes edge and prior information to guide the generation of region adjacency relationships, correcting mislabeled regions. The prior information helps weaken the impact of noise and reduces fragmented regions, while edges refine region boundaries, ensuring precise segmentation.
2) Selective label smoothing PMRF (SLS-PMRF) formulates a new label smoothing strategy to identify and eliminate the noise clusters in PMRF segmentation results, enhancing edge sharpness and reducing noise interference.
3) Equal-channel feature perturbation (EC-FP) introduces perturbations into the current features without altering their quantity, preventing performance stagnation, and helping the framework escape local optima.
4) Random probability-based feature fusion (RP-FF) randomly selects labels for each pixel position from the label maps of three features with equal probability, effectively combining all features while ensuring a single output channel.

By fully leveraging the advantages of RMRF and PMRF, ECMRF achieves improved segmentation of both regions and edges in SAR images. Additionally, perturbing filtering characteristics helps ECMRF avoid local optima, further enhancing SA. The main contributions of ECMRF are as follows.

1) We introduce multiple edge-constrained filters to enhance feature diversity and detailed richness.
2) We enhance and integrate RMRF and PMRF, using RMRF for fast and precise region partitioning while employing PMRF to sharpen corner features and suppress noise, thereby improving SAR image segmentation performance.
3) We propose a cascaded framework that combines region-level and pixel-level segmentations with feature perturbation and fusion to produce the final segmentation output, effectively eliminating internal noise while preserving detailed information.

The structure of this article is organized as follows. Section II introduces the fundamental MRF algorithms. Section III provides a detailed description of the proposed ECMRF algorithm. Section IV presents experimental images and outlines the parameter configurations required by ECMRF. Section V showcases and analyzes the experimental results. Finally, Section VI summarizes the performance of ECMRF.

II. Preliminary

Due to pixel interactions' noncausal nature and spatial proximity dependence, the PMRF, a classical MRF model, has been widely applied in image processing [27]. Given an image $I \in \mathbb{R}^{H\times W}$ and a set of labels C, the image segmentation task can be formulated as an optimization problem aimed at maximum posterior probability $P(C|I)$ [18]. According to the Bayesian theory, this optimization problem is equivalent to maximizing $p(I|C)P(C)$, which is expressed as

$$f(I,C) = \arg\max_C(P(C|I)) = \arg\max_C(p(I|C)P(C)) \quad (1)$$

where $P(C)$ denotes the prior probability. To solve this optimization problem, it is necessary to transform it into a minimization problem, with analytical representations for $p(I|C)$ and $P(C)$. This process is formulated as

$$f(I,C) = \arg\max_C\left(\varphi_f + \varphi_s\right) \quad (2)$$

where φ_f and φ_s are the analytical representations for $p(I|C)$ and $P(C)$, respectively. According to the Hammersley–Clifford theorem, the φ_s can be expressed using the energy

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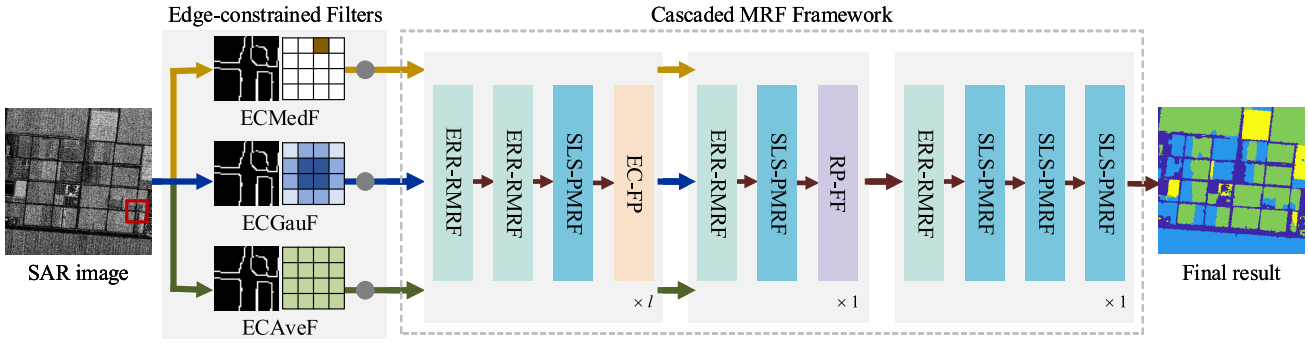


Fig. 1. Overview of our proposed ECMRF approach.

function of the Gibbs random field as follows:

$$\varphi_s = \sum_{g\in I} E_g = \beta \sum_{g\in I} \sum_{<g,t>\in Z} 1 - \delta(c_g, c_t) \tag{3}$$

where g represents a pixel in the image I, Z denotes the set of potential energy clusters for g, and E_g signifies the total potential energy of g. The pixel t is a neighbor of g, while c_g and c_t are the class labels associated with g and t, respectively. The parameter $\beta > 0$ serves as a balancing coefficient. If $c_g = c_t$, then $\delta(c_g, c_t) = 1$; otherwise, $\delta(c_g, c_t) = 0$. Clearly, φ_s effectively captures the spatial contextual information in the image.

Since the grayscale values of the pixels for each class form distinct grayscale intervals, the likelihood function $p(I|C)$ is typically represented analytically as φ_f using a Gaussian distribution, which is defined as

$$\varphi_f = \sum_{g\in I} \left\{ \frac{1}{2} \ln\left(2\pi\sigma_{c_g}^2\right) + \frac{\left(v_g - \mu_{c_g}\right)^2}{2\sigma_{c_g}^2} \right\} \tag{4}$$

where φ_f can be interpreted as a feature descriptor for the image, and v_g represents the grayscale value of pixel g. μ_{c_g} and σ_{c_g} denote the mean and standard deviation of the grayscale values for all pixels belonging to class c_g.

RMRF [18] segments an image into multiple nonoverlapping connected regions at the region level rather than pixelwise. Essentially, RMRF is a variation of PMRF designed to reduce computation time by leveraging similarities between regions while modeling contextual relationships and feature information in a similar manner to PMRF.

III. Proposed Method

This article proposes a novel ECMRF approach for unsupervised SAR image segmentation. First, we introduce multiple edge-constrained filters to extract diverse image features. Subsequently, we present a cascaded MRF framework that integrates region-level and pixel-level segmentations with feature perturbation and fusion to generate the final segmentation output. The cascaded MRF framework includes four key components: ERR-RMRF, SLS-PMRF, EC-FP, and RP-FF.

Fig. 1 illustrates an overview of the proposed ECMRF. More concretely, we first apply a series of edge-constrained filtering techniques, including ECGauF [21], edge-constrained median filtering (ECMedF), and edge-constrained average filtering (ECAveF), to enhance boundary features in the SAR image. Then, label mapping (denoted as a gray solid circle) [22] is performed on the extracted filtering features. The three resulting features, each represented as a separate channel, are then processed through the cascaded MRF framework, achieving region smoothing, edge sharpening, and corner highlighting to produce the final segmentation result.

The cascaded MRF framework, as shown in Fig. 1, consists of four key components that are flexibly integrated. It follows a hierarchical processing strategy: in the first l layers, ERR-RMRF ensures interobject smoothness, followed by SLS-PMRF for boundary sharpening, and EC-FP for feature perturbation to improve diversity. Together, these components enable progressive segmentation optimization from coarse to fine granularity. In the $(l+1)$th layer, ERR-RMRF and SLS-PMRF are retained, while RP-FF is introduced for feature fusion, maintaining output dimensions consistent with the input. In the final layer, the fused features undergo further optimization to enhance SA. Empirical evaluations recommend setting l to 4 by default. (For more details, please refer to Section IV-B.) Extensibly, if data or tasks with new requirements, the number of SLS-PMRF and ERR-RMRF can also be changed per layer.

In Sections III-A–III-E, this article provides a detailed explanation of the construction method for features with emphasized boundaries, as well as the four components within ECMRF: ERR-RMRF, SLS-PMRF, EC-FP, and RP-FF.

A. Features With Emphasized Boundaries

GA enhances population diversity, guiding search results away from local optima and toward global optimality, which can prevent algorithmic stagnation. In the context of SAR image segmentation, if filtering features are analogized to a population, the diversity of the initial population is determined by the variety of filtering methods employed. A single filtering method may yield erroneous responses that cannot be eliminated or attenuated in subsequent processing stages, ultimately hindering the improvement of SA. This scenario can also be considered as SA becoming trapped in a local optimum. Various filtering methods can generate diverse features, offering multiple options for pixel labels, thus facilitating the achievement of higher segmentation performance.

Mean, Gaussian, and median filtering are three common traditional filtering methods capable of providing mean features, Gaussian features, and median features for images. Moreover, these methods have simple theories and convenient implementation procedures. However, both the mean filter and the Gaussian filter belong to linear filters, with the former

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effectively reducing noise but causing significant edge blurring and severe detailed loss. In contrast, median filtering exhibits nonlinear response characteristics and, similar to Gaussian filtering, alleviates to some extent the adverse effects on edges and details caused by mean filtering. Facing the unique speckle noise of the SAR image, the performance of these three methods yet has limitations. We proposed an ECGauF [21], which exhibits excellent performance on SAR images by reducing speckle noise while highlighting edges. Inspired by ECGauF, this article adopts the same approach to obtain edges and introduces them as constraints into the processes of both median filtering and mean filtering, in order to enhance their filtering performance on SAR images, obtaining various features with highlighted boundaries. The edge generation formula for these edge-constrained filtering are as follows:

$$E^k = \text{canny}\left(I_s^k\right) \tag{5}$$

$$E_d^k = \text{dalitate}\left(E^k, \mathbf{I}_{3\times 3}\right) \tag{6}$$

$$E_{is} = E^{N_{\text{edge}}} \cdot \prod_{i=1}^{N_{\text{edge}}-1} \text{simplify}\left(E_d^i\right) \tag{7}$$

where canny$(\cdot)$ represents the edge detection using a Canny operator [28], and I_s^k and E^k, respectively, denote the directional Gaussian smoothing result [29] and edge detection result for the kth iteration, respectively. In (5)–(7), k, as an integer, is in the range $[1, N_{\text{edge}}]$, with N_{edge} representing the number of edge detections. dalitate$(\cdot)$ denotes the dilation operation, $\mathbf{I}_{3\times 3}$ is a 3×3 identity matrix, and E_d^k represents the edges obtained after dilating E^k using the structural element $\mathbf{I}_{3\times 3}$. simplify$(\cdot)$ is used to set nonzero values in the dilated edges E_d^k to 1 while preserving zero values. E_{is} expresses the edges used for edge-constrained filtering. Under E_{is} constraints, the formulas of Gaussian filtering, median filtering, and mean filtering are as follows:

$$GR_{x,y}^k = \text{grow}\left\{E_{is}, I_{gs}^{k-1}(x, y), W_{is}\right\}$$

$$I_{gs}^k(x, y) = \frac{1}{H} T_{\text{gau}} \cdot GR_{x,y}^k \tag{8}$$

$$AR_{x,y}^k = \text{grow}\left\{E_{is}, I_{\text{as}}^{k-1}(x, y), W_{is}\right\}$$

$$I_{\text{as}}^k(x, y) = \text{mean}\left(T_{\text{avg}} \cdot AR_{x,y}^k\right) \tag{9}$$

$$MR_{x,y}^k = \text{grow}\left\{E_{is}, I_{ms}^{k-1}(x, y), W_{is}\right\}$$

$$I_{ms}^k(x, y) = \text{median}\left(T_{\text{med}} \cdot MR_{x,y}^k\right) \tag{10}$$

where grow$(\cdot)$ represents the region-growing operation, and W_{is} is the filtering window diameter. $GR_{x,y}^k$, $AR_{x,y}^k$, and $MR_{x,y}^k$ are the results obtained after performing region growing within the filter window centered at seed points $I_{gs}^{k-1}(x, y)$, $I_{\text{as}}^{k-1}(x, y)$, and $I_{ms}^{k-1}(x, y)$, respectively. T_{avg} and H individually express the Gaussian filter template and the corresponding normalization term [29]. T_{avg} and T_{med} are both $W_{is} \times W_{is}$ unit matrices, representing the mean filtering template and median filtering template, respectively. mean$(\cdot)$ and median$(\cdot)$ separately indicate the operations of taking the mean and median. I_{gs}^k, I_{as}^k, and I_{ms}^k represent the Gaussian filtering result, mean filtering result, and median filtering result under edge constraints for the kth iteration, respectively. Specially, $I_{gs}^0 = I_{as}^0 = I_{ms}^0 = I_s^{N_{\text{edge}}}$. In (8)–(10), k in the range $[1, N_{\text{ecs}}]$, is an integer, with N_{ecs} representing the number of filtering iterations.

B. RMRF Rapidly Regulated by Edges

RMRF operates on regions as its elements, ensuring consistent pixel labels within each region and eliminating isolated noise points. The spatial relationships among regions are represented using a region adjacency graph (RAG). In the RAG, each node represents a region with a distinct label. Connections between nodes indicate adjacency and the presence of a common boundary between regions. Watershed transformation and simple linear iterative clustering (SLIC) are widely used for region generation due to their theoretical maturity; however, both methods suffer from significant drawbacks when applied to SAR images. First, they tend to result in excessively fragmented regions, leading to an increased time consumption for RMRF. Second, the obtained regions are coarse because edges are not utilized effectively, resulting in lower SA compared to PMRF. Hence, this article constructs the RAG, employing region growth on the label map with a smoothing effect to mitigate fragments and introducing edge information for region refinement. The corresponding formula is as follows:

$$G = \left\{(r,\ e) | r \in \text{grow}\left(E_{\text{deep}}, L\right)\right\} \tag{11}$$

where r and e represent the nodes and edges in G, respectively, and the RAG is generated by E_{deep} and L. E_{deep} serves as an edge map to regulate region division, and L is a label map.

In MRF algorithms, labels serve as priors. Here, labels have no physical significance and originate from unsupervised clustering. Each cluster is assigned a unique label, either a color or a number. Watershed transformation and SLIC operate directly on filtered or unfiltered SAR images, often resulting in several neighboring regions having the same label. In comparison to the SAR image, the label map derived from filtering features exhibits local consistency. Therefore, dividing regions on the label map can reduce the number of elements for RMRF, offering an acceleration. Meanwhile, edges used as constraints can refine the boundaries of regions. Even in the presence of false edges, region growth can bypass most of them because false edges are typically nonclosed.

To sum up, priors and edges can provide fewer and more refined elements to RMRF. If L_{in}, L_{out}, and E_{deep} are used to represent the input label map (prior), output label map, and input edge map, respectively, the implementation details for ERR-RMRF are listed as follows:

$$L_{\text{out}} = \arg\min_C \left(\varphi_s(L_{\text{in}},\ G) + \varphi_f(L_{\text{in}},\ I,\ G)\right) \tag{12}$$

where L_{out} is generated through RMRF, and it should be noted that the computation of the feature term φ_f and the spatial contextual term φ_s in RMRF is based on regions as fundamental elements. G is derived from (11) with inputs L_{in} and E_{deep}.

C. PMRF With Selective Label Smoothing

RMRF can effectively suppress isolated noise within the target, rectifying mismarked regions, while PMRF excels in accurately labeling edges and corners. Therefore, incorporating PMRF operations after the ERR-RMRF can ameliorate issues related to the mis-segmentation of corners and edges. Nevertheless, the inherent characteristics of speckle noise in SAR imagery are inevitably reflected to some extent in the feature model φ_f of PMRF, leading to the emergence of new

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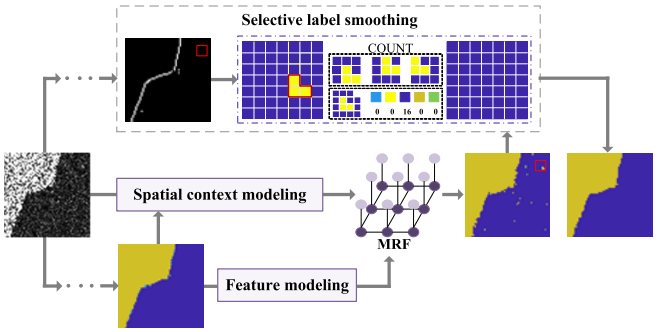


Fig. 2. Workflow of PMRF with selective label smoothing.

noise in the current segmentation results. This new noise primarily manifests within the target, appearing in isolated small clusters, wherein each noise cluster consists of one or several pixels and remains disconnected from others. Simultaneously, ERR-RMRF, influenced by factors such as the original features of SAR imagery, cannot produce a perfect label map and tends to leave some segmentation errors. Consequently, this article proposes a selective label smoothing approach to mitigate the noise impact introduced by PMRF operations. It offers the potential for rectifying the errors left by ERR-RMRF, with the correspondingly specific steps as follows.

1) Region division is performed on the label map using region growth, and regions with sizes smaller than the maximum noise size, denoted as S_{nr}, are identified and recorded as a set of noise clusters.
2) For any noise point within a noise cluster, if that noise point is not an edge point, a region-growing operation, constrained by the edges, is carried out within a smoothing window with a diameter of W_{ls}, centered at that noise point. Otherwise, no operation is performed on that noise point.
3) For all noise points within a noise cluster, a candidate label set is composed of labels from pixels, excluding the noise point itself, within the growth region of each noise point. It is important to note that there may be overlaps in the growth regions, and labels of pixels in the overlap regions are counted only once.
4) For any cluster within the noise cluster set, a count for the occurrences of each type of label in its candidate label set is conducted, and the label with the highest occurrence count is assigned as the label for that cluster.

To ensure precise labeling for corners and edges without introducing new noise, and to further rectify the mislabeled regions left by ERR-RMRF, this article applies selective label smoothing to the label map produced by PMRF, constructing the SLS-PMRF, as visually described in Fig. 2.

In Fig. 2, the input label map serves as prior information to construct a spatial context model φ_s and, when combined with SAR imagery, forms the feature model φ_f. The output of the MRF module, after selective label smoothing, yields the output label map of the SLS-PMRF without additional noise cluster and edge errors. It is important to note that the output of the MRF in Fig. 2 represents the results after N_{iter} iterations.

D. Equal-Channel Feature Perturbation

In order to add upward momentum to stagnant performance, this article introduces perturbations into the current features without changing the quantity, thus proposing an EC-FP strategy. It involves pairwise fusion of input channels to maintain the number of input label maps matches the number of output label maps. Assuming the input label maps for a certain EC-FP are $\{L^{i1}, L^{i2}, L^{i3}\}$, and the corresponding edge maps are $\{E^{i1}, E^{i2}, E^{i3}\}$, while the output label maps are $\{L^{o1}, L^{o2}, L^{o3}\}$. L^{o1}, L^{o2}, and L^{o3} are obtained by fusing L^{i1} with L^{i2}, L^{i2} with L^{i3}, and L^{i3} with L^{i1}, respectively. The implementation details of EC-FP are shown in Algorithm 1.

Algorithm 1 EC-FP

Input: $\{L^{i1}, L^{i2}, L^{i3}\}$ and $\{E^{i1}, E^{i2}, E^{i3}\}$
Output: $\{L^{o1}, L^{o2}, L^{o3}\}$
for (u, v) in $\{(1, 2), (2, 3), (3, 1)\}$ **do**
  for $x = 1 : H$ **do**
    for $y = 1 : W$ **do**
      if $L^{iu}(x, y) = L^{iv}(x, y)$ **then**
        $L^{ou}(x, y) = L^{iu}(x, y)$
      else
        $R^1 = grow\{L^{iu}(x, y), E^{iu}, W_{fp}\}$
        $R^2 = grow\{L^{iv}(x, y), E^{iv}, W_{fp}\}$
        $L^{ou}(x, y) = most_and_count(R^1 \cup R^2)$
      end if
    end for
  end for
end for

Algorithm 2 Feature Fusion Based on Random Probability

Input: $\{L_f^{i1}, L_f^{i2}, L_f^{i3}\}$
Output: L_f^o
for $x = 1 : H$ **do**
  for $y = 1 : W$ **do**
    $a = rand()$ ▷ $a \in [0, 1)$ is a random probability
    if $0 \le a < 1/3$ **then**
      $L_f^o(x, y) = L_f^{i1}(x, y)$
    else if $1/3 \le a < 2/3$ **then**
      $L_f^o(x, y) = L_f^{i2}(x, y)$
    else
      $L_f^o(x, y) = L_f^{i3}(x, y)$
    end if
  end for
end for

The label maps have the same shape as SAR image I, which is $H \times W$. The function most_and_count$(\cdot)$ can count the labels of all pixels within the growth regions R^1 and R^2, and select the label with the highest pixel count to assign to the pixel point (x, y).

E. Feature Fusion Based on Random Probability

After repeated and cascaded uses of ERR-RMRF, SLS-PMRF, and EC-FP, the label maps for each channel have reached a relatively stable state, where further perturbations are unlikely to yield significant positive effects on performance. Given the current approximate SA across channels and the ultimate output being a single channel, this article proposes a feature fusion strategy based on random probability. Assuming the input label maps for the RP-FF are $\{L_f^{i1}, L_f^{i2}, L_f^{i3}\}$, the corresponding calculating details for the output label map L_f^o are shown in Algorithm 2.

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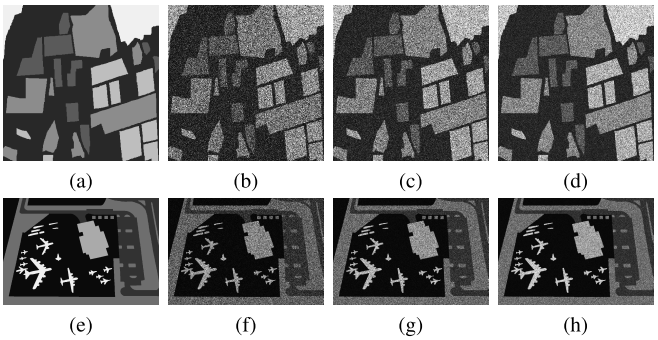


Fig. 3. Simulated SAR images: (a) SSI1-GT; (b)–(d) one-, two-, and four-look SSI1; (e) SSI2-GT; and (f)–(h) one-, two-, and four-look SSI2.

IV. Data and Parameters

Segmentation performance will be investigated on two sets of simulated SAR images and one set of real SAR images, and key parameters will be introduced, for ECMRF. It should be noted that the real SAR images used are grayscale images after multiple corrections, not raw data.

A. Experimental Data

Adding varying levels of multiplicative noise [30] to two different clean images results in two sets of simulated SAR images,[1] as depicted in Fig. 3.

As shown in Fig. 3(a) and (e), SSI1-GT and SSI2-GT are clean and noise-free. Serving as ground truths, they are each subjected to three levels of noise to obtain the image sets SSI1 [as shown in Fig. 3(b)–(d)] and SSI2 [as shown in Fig. 3(f)–(h)]. The imagery look is used to measure the noise intensity of the image, where a lower look corresponds to higher noise, and vice versa. Consequently, the sizes and target types of the noisy images match those of their corresponding ground truths. SSI1-GT has equal length and width, totaling 460 × 460 pixels, while SSI2-GT has unequal dimensions, totaling 800 × 545 pixels. Both SSI1-GT and SSI2-GT contain five target types, including numerous small and irregular targets, with corresponding gray values of {40, 90, 140, 190, 240} and {10, 70, 130, 180, 240}, respectively. Fig. 3 shows that the data have diverse noise intensities, including odd-look, even-look, high-look, and low-look images. In summary, the SSI1 and SSI2 image sets serve as a testbed for evaluating the segmentation capability of algorithms with respect to high and low noises, regular and irregular targets, large and small sizes, as well as simple and complex structures.

The three real SAR images, named according to their capture locations, are depicted in Fig. 4.

As shown in Fig. 4, the real SAR images contain various land objects, including ponds, rivers, buildings, roads, forests, grasslands, and various crops, among others. The resolutions of the three SAR images are 1 m, with dimensions from left to right as follows: 350 × 350, 256 × 256, and 831 × 1401. Both Maricopa and Xi'an are smaller in size and can be segmented using four types of labels. However, Maricopa contains more small objects, while Xi'an exhibits higher noise levels. In contrast, Rosenheim is larger in size.

[1]https://github.com/JJiangLi/simulated-SAR-images

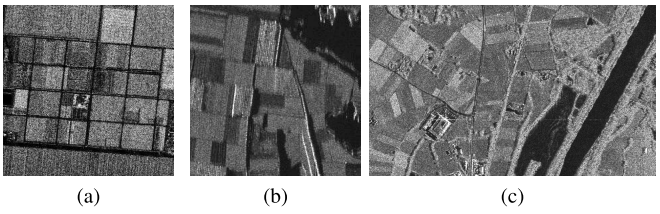


Fig. 4. Real SAR images: (a) Maricopa; (b) Xi'an; and (c) Rosenheim.

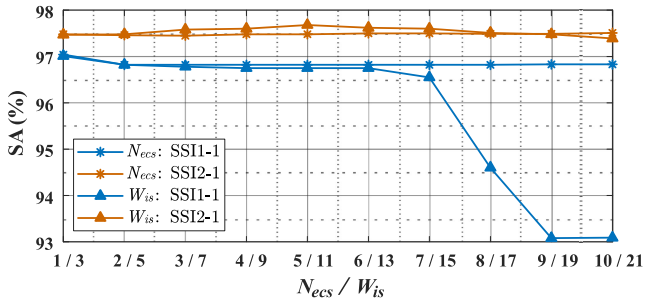


Fig. 5. Impact of N_{ecs} and W_{is} on SA.

Rosenheim can be segmented using four types of labels and features a higher number of irregular objects. These real SAR images, with different sizes, numbers of label types, complexity levels, target shapes, and noise levels, provide a comprehensive testing ground for evaluating the segmentation performance of algorithms.

B. Parameter Configuration

The parameters involved in ECMRF primarily include K for K-means, filtering times N_{ecs} and filtering window diameter W_{is} required for feature generation, iteration times N_{iter}, smoothing window diameter W_{ls}, and maximum noise cluster size S_{nr} required for SLS-PMRF, and sliding window diameter W_{ef} required for EC-FP. While ensuring SA, this article simplifies the algorithm by fixing certain parameter values. Specifically, K is preset according to the input image. $W_{ls} = 15$ and $W_{ef} = 5$. The values of N_{iter} for eight SLS-PMRFs are taken from one of the following sequences: {6, 3, 6, 6, 1, 1, 1, 1}, {3, 1, 3, 3, 1, 1, 1, 1}, {1, 1, 1, 1, 1, 1, 1, 1}, or {1, 0, 0, 0, 0, 0, 0, 1}, with a choice made based on the characteristics of the SAR images. S_{nr} maintains consistent values at each layer; concretely, $S_{nr} = \{30, 30, 30, S_{nr}^4, 20, 20\}$, where S_{nr}^4 represents the maximum noise cluster size for the fourth layer of ECMRF. This parameter is identical to N_{ecs} and W_{is} and is determined by the experimental SAR images. Next, the sensitivity analysis on N_{ecs} and W_{is} is conducted. Specifically, the impact of N_{ecs} (conditions: $W_{is} = 5$ and $N_{\text{ecs}} \in \{1, 2, \ldots, 10\}$) and W_{is} (conditions: $N_{\text{ecs}} = 4$ and $W_{is} \in \{3, 5, \ldots, 21\}$) on SA is tested using SSI1-1 and SSI2-1, as shown in Fig. 5.

As shown in Fig. 5, under the condition of $W_{is} = 5$, N_{ecs} has a relatively small impact on SA. Notably, the yellow star line corresponding to SSI2-1 tends to remain flat. Therefore, N_{ecs} can take any integer in $[1, 10]$. In contrast, SSI2-1 is not sensitive to changes in W_{is}. However, when W_{is} is larger, the SA of SSI1-1 decreases rapidly. This is mainly because many of the targets in SSI1-1 are small in size, and excessive W_{is}

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TABLE I
NUMERICAL SEGMENTATION RESULTS ON SIMULATED SAR IMAGES

Algorithm	1-look SSI1		2-look SSI1		4-look SSI1		1-look SSI2		2-look SSI2		4-look SSI2	
	SA%	KC	SA%	KC	SA%	KC	SA%	KC	SA%	KC	SA%	KC
THFCM [31]	85.36	0.789	90.84	0.866	94.09	0.913	93.91	0.911	94.34	0.918	94.65	0.923
CSHLC [21]	94.33	0.914	97.91	0.969	99.16	0.987	93.68	0.909	98.65	0.980	99.18	0.988
FRCBLP [32]	97.18	0.958	97.50	0.963	97.60	0.964	96.85	0.954	97.50	0.964	97.63	0.966
RMTPA [33]	80.58	0.695	88.50	0.828	97.74	0.966	92.05	0.883	97.27	0.960	97.93	0.970
FWMRF [20]	59.73	0.419	66.63	0.522	77.69	0.678	73.38	0.627	81.08	0.734	86.57	0.807
RCC-MRF [18]	93.34	0.902	94.04	0.912	95.30	0.931	95.29	0.932	95.79	0.939	96.28	0.946
ECMRF	**97.84**	**0.968**	**98.64**	**0.980**	**99.28**	**0.989**	**98.29**	**0.975**	**99.13**	**0.987**	**99.57**	**0.994**

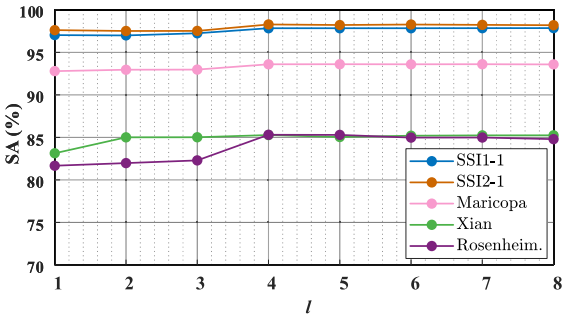


Fig. 6. Impact of l on SA.

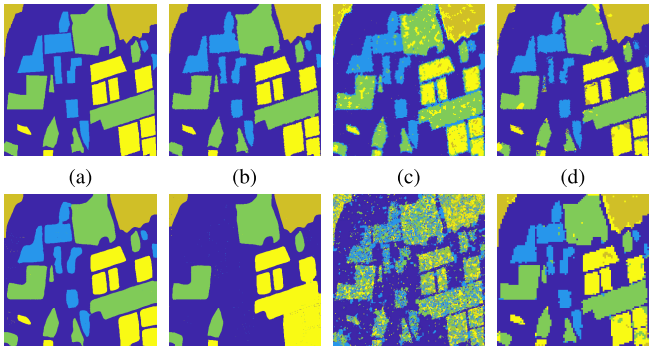


Fig. 7. Graphical segmentation results on one-look SSI1: (a) ground truth; (b) ECMRF; (c) THFCM [31]; (d) CSHLC [21]; (e) FRCBLP [32]; (f) PMTPA [33]; (g) FWMRF [20]; and (h) RCC-MRF [18].

can damage the structure of the small targets after filtering. Considering all factors, when $N_{\text{ecs}} = 4$, W_{is} can be set to values such as 3, 5, 7, 9, 11, or 13.

Meanwhile, l is also a crucial parameter for the cascaded MRF formwork. Fig. 6 showcases the overall performance change following the increasing l.

As shown in Fig. 6, when $l \geq 2$, the SA of the Xi'an image remains most stable and optimal. The other images achieve their optimal at $l = 4$ and subsequently exhibit only minor fluctuations as l increases. Therefore, to balance accuracy and efficiency, we set l to 4 by default.

V. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

The segmentation performance in SAR images is evaluated for ECMRF, THFCM [31], CSHLC [21], FRCBLP [32], RMTPA [33], FWMRF [20], and RCC-MRF [18], using SA and kappa coefficient (KC) as metrics. It is worth noting that RCC-MRF is supervised, while the others are unsupervised. THFCM and CSHLC obtain initial labels using FCM and K-means, respectively. FRCBLP, RMTPA, and RCC-MRF perform final segmentation with a focus on regions rather than individual pixels. Both CSHLC and FWMRF employ PMRF, with the latter also utilizing RMRF, while RCC-MRF exclusively employs RMRF. ECMRF operates in an unsupervised manner, and for fairness in comparison, RCC-MRF is trained with a limited number of annotated data. In addition, the three improved edge-preserving filtering methods and the running time of these algorithms are discussed.

A. Results and Analysis of Simulated SAR Images

With the SSI1 and SSI2 image sets as simulated segmentation images, the KC and SA for each algorithm were calculated, as shown in Table I. Concerning the simulated SAR image sets SSI1 and SSI2, FWMRF exhibits the poorest segmentation performance, primarily manifested by lower SA and higher noise sensitivity. In terms of numerical values, the segmentation performance of THFCM is noticeably better than that of FWMRF; however, it still falls short of the performance achieved by the other algorithms. FRCBLP shows the smallest differences in numerical results when segmenting images with varying degrees of contamination in the same dataset, demonstrating stable performance under the influence of noise, but PMTPA presents contradictory findings. Under the condition of limited annotated training samples, RCC-MRF produces numerical results lower than FRCBLP, and noise variations have a relatively minor impact on their performance.

As shown in Table I, CSHLC achieves numerical results higher than FRCBLP on two-look and four-look simulated SAR images, and for four-look images, in terms of SA and KC, it even approaches the results of ECMRF. However, CSHLC is not well-suited for high-noise images, with significantly lower SA values when segmenting one-look SSI1 and one-look SSI2 compared to FRCBLP. In contrast, ECMRF obtains the highest SA and KC under three noise levels in both datasets, indicating that the proposed algorithm can stably and accurately segment simulated SAR images.

Fig. 7 shows the graphical segmentation results of each algorithm for one-look SSI1.

As shown in Fig. 7, CSHLC and RCC-MRF exhibit similar graphical segmentation results for one-look SSI1, with their corresponding SA and KC being close to one another. Both algorithms produce numerous small clusters of misclassified pixels within the target regions. The result of THFCM demonstrates limitations in accurately segmenting both edge pixels

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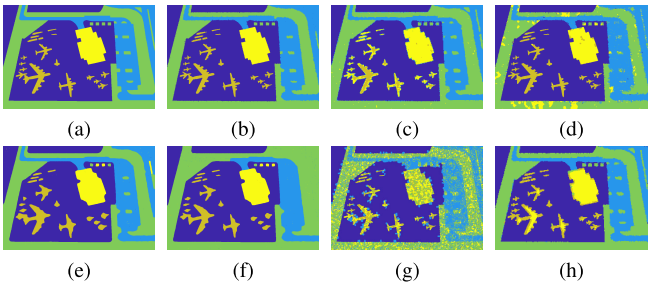


Fig. 8. Graphical segmentation results on one-look SSI2: (a) ground truth; (b) ECMRF; (c) THFCM [31]; (d) CSHLC [21]; (e) FRCBLP [32]; (f) PMTPA [33]; (g) FWMRF [20]; and (h) RCC-MRF [18].

TABLE II
NUMERICAL SEGMENTATION RESULTS ON REAL SAR IMAGES

Algorithm	Maricopa		Xian		Rosenheim	
	SA%	KC	SA%	KC	SA%	KC
THFCM [31]	80.89	0.720	81.37	0.734	73.44	0.628
CSHLC [21]	91.43	0.876	82.79	0.752	79.61	0.713
FRCBLP [32]	88.73	0.837	79.50	0.705	81.23	0.736
RMTPA [33]	76.91	0.659	74.49	0.623	77.58	0.683
FWMRF [20]	73.19	0.618	66.77	0.529	69.37	0.555
RCC-MRF [18]	85.12	0.784	80.24	0.722	75.09	0.645
ECMRF	**93.60**	**0.908**	**85.28**	**0.791**	**85.31**	**0.789**

and internal target pixels. Consequently, the latter yields lower SA and KC compared to the former two. FWMRF fails to distinguish between the green and brown targets within SSI1, and RMTPA integrates the light blue targets into the dark blue ones, so their graphical results differ significantly from the ground truth in terms of the color-label distribution. Both ECMRF and FRCBLP yield graphical results without noticeable noise points, but ECMRF, the proposed algorithm, better preserves the corner information in SSI1.

Fig. 8 shows the graphical segmentation results of each algorithm for one-look SSI2.

As depicted in Fig. 8, FWMRF confuses pixels marked green and yellow in SSI2, primarily because its segmentation performance is imperfect, rendering it unable to effectively differentiate between two target types with similar grayscale values. THFCM and RCC-MRF erroneously label a significant number of pixels within and around targets shaped like airplanes. Additionally, CSHLC incorrectly delineates larger yellow regions inside green targets. And, RMTPA fails to recognize the small green targets on the right side of the SSI2. Among the compared algorithms, FRCBLP evidently achieves the most favorable graphical results.

However, the presence of mislabeled regions still limits its KC, and the overly smoothed corners result in coarser target contours, as shown in Fig. 8(e). In contrast, ECMRF not only refines the edges to highlight the target outlines, but also exhibits clean graphical results devoid of cluttered noise points.

B. Results and Analysis of Real SAR Images

With Maricopa, Xi'an, and Rosenheim as real segmentation images, the KC and SA for each algorithm are computed, as shown in Table II.

As shown in Table II, when considering the overall performance, RCC-MRF outperforms FWMRF in segmenting

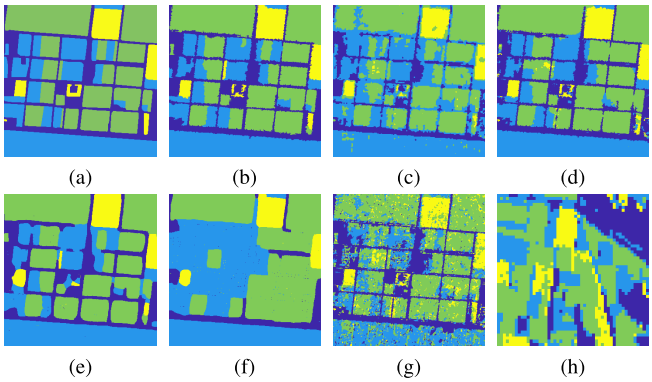


Fig. 9. Graphical segmentation results on Maricopa: (a) ground truth; (b) ECMRF; (c) THFCM [31]; (d) CSHLC [21]; (e) FRCBLP [32]; (f) PMTPA [33]; (g) FWMRF [20]; and (h) RCC-MRF [18].

real SAR images. Both utilize MRF, but the former is supervised. Combining the information from Table I, it is evident that FWMRF exhibits the poorest segmentation performance. RCC-CNN and THFCM produce higher numerical evaluations for Xi'an compared to FRCBLP, but for the remaining real images, they yield lower SA and KC. RMTPA performs worse than FRCBLP on real SAR images but is superior to RCC-CNN and THFCM for Rosenheim. For Maricopa, with numerous small targets, and Xi'an, with substantial noise, CSHLC achieves the second-highest SA and KC although they remain significantly lower than those of ECMRF. ECMRF and FRCBLP segment the Rosenheim image that is characterized by larger sizes, obtaining SAs of 85.31% and 81.23%, respectively, with the former surpassing the latter by 4.08%. And, the remaining algorithms yield SAs below 80%. In summary, ECMRF demonstrates superior segmentation performance when applied to real SAR images.

Fig. 9 shows the graphical segmentation results of each algorithm for the image, Maricopa.

As depicted in Fig. 9, THFCM fails to refine the edges, resulting in the incorrect segmentation for a large number of edge pixels. FRCBLP does not sharpen the corners, and the target outlines in its segmented image appear overly smooth. In contrast, although FWMRF exhibits poorer segmentation performance, it effectively leverages the advantages of PMRF, resulting in well-defined corners and edges in the image. Compared to RCC-MRF, CSHLC produces segmented edges that are smoother. PMTPA fails to extract boundaries, resulting in multiple targets being mixed together. Among the compared algorithms, the segmentation result of CSHLC appears visually advantageous, but it contains several mislabeled pixel clusters, resulting in a somewhat cluttered graphical segmentation when compared to ECMRF. Evidently, ECMRF successfully preserves small targets in the image and achieves a cleaner segmentation result, benefiting from the multiple uses of PMRF.

Fig. 10 shows the graphical segmentation results of each algorithm for the image, Xi'an.

As shown in Fig. 10, THFCM and RCC-MRF exhibit limited segmentation capability for edge pixels, and the consistency of target interior pixels in their graphical segmentation results is low. PMTPA cannot effectively identify the yellow targets in the image. CSHLC achieves the second-highest SA for processing Xi'an image, as depicted in Table II,

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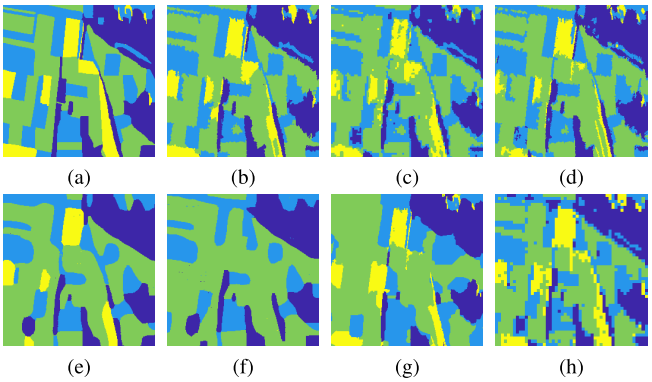


Fig. 10. Graphical segmentation results on Xi'an: (a) ground truth; (b) ECMRF; (c) THFCM [31]; (d) CSHLC [21]; (e) FRCBLP [32]; (f) PMTPA [33]; (g) FWMRF [20]; and (h) RCC-MRF [18].

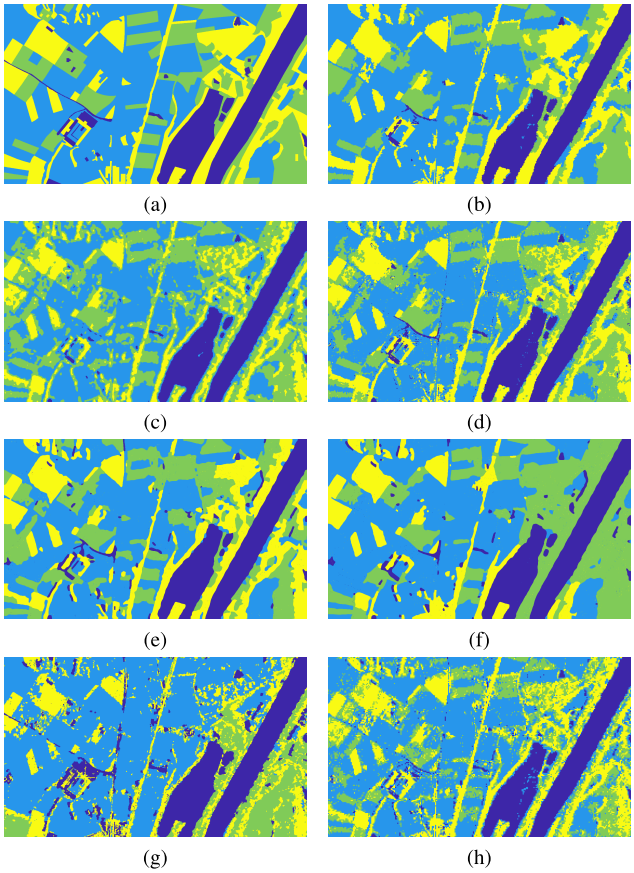


Fig. 11. Graphical segmentation results on Rosenheim: (a) ground truth; (b) ECMRF; (c) THFCM [31]; (d) CSHLC [21]; (e) FRCBLP [32]; (f) PMTPA [33]; (g) FWMRF [20]; and (h) RCC-MRF [18].

but its graphical result features edges that have not been adequately smoothed. FRCBLP and FWMRF obtain relatively smooth segmented edges; however, they face difficulties in segmenting some small targets. The primary reason is that neither of them considers the preservation and utilization in edge information. In contrast, ECMRF introduces edge information and effectively combines the advantages of RMRF and PMRF, thereby achieving the best segmentation performance.

Fig. 11 shows the graphical segmentation results of each algorithm for the image, Rosenheim.

TABLE III
NUMERICAL EVALUATION FOR THE IMPACT OF FILTERS

Algorithm	Maricopa		Xian		Rosenheim	
	SA%	KC	SA%	KC	SA%	KC
ECMRF-ECGauF	90.94	0.868	79.56	0.702	79.78	0.709
ECMRF-ECMedF	90.60	0.864	82.84	0.755	79.83	0.709
ECMRF-ECAveF	90.21	0.859	80.06	0.711	80.59	0.720
ECMRF	**93.60**	**0.908**	**85.28**	**0.791**	**85.31**	**0.789**

The graphical result presented in Fig. 11(d) is visually closer to the ground truth compared to the one in Fig. 11(h), and the corresponding SA and KC also provide evidence. This suggests that the unsupervised CSHLC demonstrates segmentation capabilities on par with the supervised RCC-MRF when dealing with a limited number of annotated samples. Furthermore, similar to THFCM, the label distributions in their respective graphical results appear relatively chaotic, as all three algorithms exhibit pronounced mis-segmentation both along the image boundaries and within the target regions. FWMRF fails to effectively segment the green targets on the left side of the image, resulting in a profusion of erroneously labeled pixels as light blue in its graphical result. In contrast, FRCBLP and PMTPA, with regions as segmentation units, prevent the occurrence of isolated noise points, ensuring the cleanliness of its graphical results. However, PMTPA with inadequate segmentation capability for small targets results in the loss of many edges. The graphical result of ECMRF shares similar characteristics with FRCBLP, but the former boasts more precise edges, making the proposed algorithm more competitive.

C. Impact of Three Filtering Methods

To assess the impact of three edge-preserving filtering methods on overall performance, ECGauF, ECMedF, and ECAveF are individually integrated into ECMRF, forming three new methods: ECMRF-ECGauF, ECMRF-ECMedF, and ECMRF-ECAveF. The segmentation results of these approaches, along with ECMRF, on real SAR images are presented in Table III.

From Table III, it is evident that for the Maricopa, the segmentation performance follows this order: ECMRF outperforms ECMRF-ECGauF, which surpasses ECMRF-ECMedF, and finally ECMRF-ECAveF. For the Rosenheim, the order is slightly different: ECMRF achieves the best performance, followed by ECMRF-ECGauF, ECMRF-ECMedF, and ECMRF-ECAveF. However, similar to Maricopa, the numerical differences in segmentation performance on Rosenheim among the three new frameworks are not significant. This suggests that the impact of the three filters on Rosenheim is comparable. In contrast, the segmentation results for Xi'an show that ECMRF-ECMedF performs significantly better than ECMRF-ECGauF and ECMRF-ECAveF, indicating that ECMedF has the most positive impact on Xi'an within the ECMRF. Nevertheless, the segmentation performance of ECMRF with one of these filters is considerably lower than that with the combined three filters.

D. Running Time

One-look images from the SSI1 and SSI2 datasets, along with real SAR images, are selected as experimental data,

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TABLE IV
RUNNING TIME ON SAR IMAGES

Image	THFCM [31]	CSHLC [21]	FRCBLP [32]	PMTPA [33]	FWMRF [20]	RCC-MRF [18]	ECMRF
	Time(s)	Time(s)	Time(s)	Time(s)	Time(s)	Time(s)	Time(s)
1-look SSI1	25.47	5.49	13.39	181.95	14.69	53.03	26.89
1-look SSI2	66.89	6.14	39.82	415.68	28.00	441.55	213.29
Maricopa	3.62	1.72	4.99	87.32	12.10	57.03	19.30
Xian	1.76	1.89	2.43	42.96	10.45	16.84	5.31
Rosenheim	121.86	33.98	626.95	1271.04	85.05	1024.01	664.12

ensuring diversity in size and noise intensity. The running time of each algorithm is compared, and the corresponding numerical results are presented in Table IV.

As shown in Table IV, larger images require longer running times, while smaller images result in shorter ones, meaning that the size of the image significantly impacts the running time. RMTPA and RCC-MRF are slower, primarily because the former involves repeated calculations of fusion costs, whereas the latter requires training extensive network parameters. RMTPA builds upon and improves FRCBLP; however, FRCBLP employs a simpler fusion cost formula, making it faster. The running times of CSHLC, THFCM, and FWMRF are relatively short, with CSHLC being the most efficient. While ECMRF has a longer running time due to its cascaded segmentation framework, it is still significantly faster than both RMTPA and RCC-MRF.

VI. CONCLUSION

To fully exploit the strengths of MRF for SAR image segmentation, we propose an ECMRF approach, designed to simultaneously enhance both region and edge segmentation for improved overall accuracy. We first introduce multiple edge-constrained filters to increase feature diversity and enrich detailed richness. Subsequently, we present a cascaded MRF framework that sequentially incorporates region-level and PMRF, along with feature perturbation and feature fusion operations. This framework ensures local region label consistency while preserving edge information, ultimately maximizing the accuracy of segmented targets.

This article evaluates the segmentation performance of ECMRF alongside six state-of-the-art algorithms on both simulated and real SAR images with varying noise levels and target sizes. The graphical experimental results clearly demonstrate that ECMRF exhibits an absolute advantage in noise suppression and edge sharpening. In the case of simulated SAR images, ECMRF outperforms the other algorithms, achieving SAs consistently above 97.5% and KCs exceeding 0.965, while demonstrating strong resilience to noise interference. When it comes to real SAR images with substantial content and noise variations, the evaluation metrics for ECMRF also significantly surpass those of the compared algorithms.

ECMRF achieves competitive results in terms of SA and KC, effectively emphasizing target contours and sharpening corner points. The independent four key components enhance the flexibility and adaptability of the cascaded framework. However, its segmentation capability for small targets in high-noise SAR images needs further improvement. In future work, we will consider incorporating additional segmentation factors and attempting to absorb the mechanisms of deep learning, to improve the performance of unsupervised algorithms on SAR images, especially in the context of high-noise SAR images.

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